Reviewer Recommendation and Comments for Manuscript Number ICMX-D-26-00143

Unbiased miRNA biomarker discovery for Detection of Severe Sepsis

Original Submission
Willem Buys

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Recommendation: Major Revision

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Level of interest Please indicate how interesting you found the manuscript:	An article of limited interest
Quality of written English Please indicate the quality of language in the manuscript:	Acceptable
Declaration of competing interests Please complete a declaration of competing interests, considering the following questions: 1. Have you in the past five years received reimbursements, fees, funding, or salary from an organisation that may in any way gain or lose financially from the publication of this manuscript, either now or in the future? 2. Do you hold any stocks or shares in an organisation that may in any way gain or lose financially from the publication of this manuscript, either now or in the future? 3. Do you hold or are you currently applying for any patents relating to the content of the manuscript? 4. Have you received reimbursements, fees, funding, or salary from an organization that holds or has applied for patents relating to the content of the manuscript? 5. Do you have any other financial competing interests? 6. Do you have any non-financial competing interests in relation to this paper? If you can answer no to all of the above, write 'I declare that I have no competing interests' below. If your reply is yes to any, please give details below.	I declare that I have no competing interests.

Reviewer Comments to Author

Reviewer Summary:
In this observational study entitled "Unbiased miRNA biomarker discovery for Detection of Severe Sepsis" Loh et al. quantify a set of pre-selected miRNAs in patients with neurological hemorrhage and with or without sepsis at three timepoints. The rationale for this study is strong and the novelty is moderate. Collecting structured longitudinal data from >100 sepsis patients is challenging and these data must ultimately be published. Unfortunately, in its current form, the manuscript has a number of critical flaws, as detailed in the itemized comments below. Most importantly, the inclusion criteria, sampling strategy, and selection of controls are currently quite vague and sometimes contradictory. The manuscript does not detail the adherence to appropriate reporting guidelines, for example STROBE-ME. This study is also missing any mechanistic confirmation or external validation of their findings. Data analysis is relatively shallow given the immense wealth of data the authors collected and the discussion is too unspecific to interpret the main findings of this study by proxy instead.

Recommendation: Major Revisions.
After addressing all comments below, this study will make for a decent concise report. However, I urge the authors to instead search for deeper and biologically meaningful correlations within their impressive dataset, confirm and validate key findings, and thereby elevate this publication from a mediocre report with an impressive dataset to an impressive article altogether.

Major Comments:
1. Which reporting guideline was used to prepare this manuscript? Please choose an appropriate reporting guideline, likely from among the various STROBE guidelines, and report your data according to that guideline. Reporting according to established guidelines is critical for compatibility with later meta-reviews. If you did already prepare this study according to an appropriate reporting guideline, please state the guideline used in your methods section and consider attaching the checklist as a supplemental file.
2. The sampling criteria and timing used in this study are currently unclear and somewhat contradictory. Please clarify the inclusion criteria, sampling strategy, and patient categorization throughout the manuscript.
- Methods, line 123: Sampling in relation to fever spikes is a little outdated but has biological rationale. I commend the authors for stating this so clearly in the abstract. How were sepsis cases without fever handled, which are common especially in the elderly population? I recommend also including documentation on how your blood draw timings relate to the time of sepsis diagnosis by Sepsis-3 criteria, first antibiotic use, and first collection of blood cultures in a given patient, as opposed to the first fever spike.
- Methods, lines 101-105 & line 128: According to related literature, about 30% of patients with cerebral hemorrhage develop central fever. Please state clearly whether all patients in this study, including controls, had a fever spike. If yes, I commend the authors on using such stringent controls but highly recommend also including a baseline control of Neuro-ICU patients with cerebral hemorrhage but without any signs of infection, including fever. If control patients had no fever spike, how was sample timing decided in the control group?
- Results, line 197-199: The method section had suggested that patients were sampled on days 1, 3, and 5 *after* an initial fever spike. This section suggests that sepsis patients were included in the study using some unclear other criteria and that fever spikes occurred *during the observation period*. This unclear definition of the terms *early-onset* and *late-onset* sepsis hampers accurate interpretation, precluding review.
- The methods section details the collection of ample clinical and demographic data from patients and controls. Please present these aggregate data, for example as table 1, to characterize your cohort. If your ethics review and patient consent allows, please strongly consider presenting k-anonymous data in a safe harbor repository to enable secondary analyses.

Minor Comments:
3. Title: The term "severe sepsis" should not be used anymore according to the Sepsis-3 guidelines, as it is not well-defined.
4. Introduction, lines 82-83: The introduction is refreshingly concise. However, in its current form, the research question of this study does not follow naturally from the introduction. The authors broadly introduce miRNA research in sepsis. I would recommend including some prominent examples of well established miRNA-pathology couplings beyond name-dropping certain miRNAs. I would also recommend clearly stating the research question of this study that has not yet been answered by similar previous studies.
5. Methods, line 94f: "neutropenia", "immunocompromised state", and "other factors that might affect study outcome" are very vague and do not allow reproducing this study. Please provide a clear, actionable list of diagnoses and medications that led to study exclusion. Such a list is commonly found in the supplement of similar studies.
6. Methods, line 96: Did legally authorized representatives provide *written* informed consent or was verbal informed consent considered sufficient for study inclusion?
7. Methods: Limiting study inclusion to "patients with neurological hemorrhage" may be a legitimate departmental constraint. However, this constraint to a very specific patient population may limit generalizability and should thus be included in the manuscript title, e.g., "in a cohort of 116 ICU patients with neurological hemorrhage" or similar. Study reporting guidelines (see above) usually include guidance on informative, discovery-optimized titles.
8. Methods, line 114: Which type of mortality was documented? Hospital mortality or also 7 day mortality, 28 day mortality, etc., even after discharge? How was loss-to-follow up handled? No information on mortality, e.g., as informed by findings from miRNA analyses, is presented in the manuscript.
9. Methods, line 129: These numbers add up to 60, not to 116, as stated in the abstract. Which number is correct?
10. Methods, line 136-137: How did the authors decide which miRNA's to include in their panel? In future studies on the remaining material, it may be interesting to use sequencing methods instead to avoid limiting discovery to known targets of interest.
11. Methods, general: Please consider generally adding catalog numbers, either in the text or as a supplemental materials table.
12. Conclusion: The authors suggest that these markers could be used for early sepsis detection. However, no sampling before fever spikes is being reported. Accordingly, this is a hypothesis that should be clearly worded as an outlook requiring further studies.
13. Line 201ff: Please introduce or thoroughly discuss the somewhat arbitrary definition of late vs. early-onset sepsis depending on the timing of a fever spike after some nebulous starting point. This is not standard nomenclature except in sepsis of the newborn. Do you mean early vs late sepsis *phases* or *stages*, not early vs late ~~onset~~ sepsis? Please also discussing septic hyper- and hypo-inflammation with more nuance: The two-stage model of sepsis as early hyperinflammation and late hypo-inflammation / immune paralysis is quite outdated. Instead, features of hyperinflammation and hypoinflammation seem to occur in parallel throughout sepsis pathology, even within the first 24h of diagnosis. Many sections of the discussion, e.g., lines 343ff & 352ff hinge on this flawed assumption. Such broad assumptions are especially problematic when they are not accompanied by any mechanistic data.
14. Line 222f: Please formulate clearly what you considered as a significant dysregulation, e.g., "showed no statistically significant difference between patients with and without sepsis irrespective of SIRS (|Log2-fold difference| >1, q<0.05)." (if this was the chosen comparison). The same applies to the equivalent day 3 and day 5 sections, too.
15. Lines 227ff: Please consider denoting these as q-values if they have been multiplicity adjusted. Due to the highly parallel nature of these tests, these must be multiplicity corrected if they have not been already. Please also add table descriptions.
16. The methods section suggests that patients were stratified by sepsis status, SIRS, and septic shock. This data is not shown. It may make sense to also show additional analyses stratified by age, sex, and comorbidity (e.g., Charleson Comorbidity Index).
17. Line 316ff: Positive correlation can also indicate a regulatory function to preserve homeostasis (rheostat). I recommend introducing previous work on the miRNAs discovered in this study with a little more detail and to be more specific when interpreting the concrete findings of this study rather than summoning broad concepts like "inflammation" or "apoptosis".
18. Lines 320-323 and corresponding results section: It is important to check findings of GO analyses for biological relevance. For example, most GO analyses show housekeeping pathways with little interpretable relevance to the pathology of interest.

19. Line 337, "consistent": No data specifically showing the consistency across the dataset, e.g., phase separation of septic and non-septic patients regarding this miRNA, is presented in the corresponding results section.

Language & Proofing:
20. This may be a function of the editorial manager, but there are currently two different line numberings displayed in the reviewer copy. One of them doesn't match the actual lines, so comments are made using the coherent line numbering.
21. Line 197 & 212: Readers will likely be familiar with heatmaps.
22. Lines 275f: This sentence is currently unclear, especially "in a sample of 13".
23. The available figure quality is insufficient for an accurate review. Please communicate with the editor whether the original figure quality was insufficient, whether the file format was inappropriate, or whether the figure quality was automatically downscaled in the editorial manager after your submission.
24. Line 406: I assume the authors mean that they did not receive *external* funding for this study, i.e., the study was funded from institutional resources?

Further Suggestions:
25. Currently, the results section contains quite some interpretation and discussion in the context of related literature. You might consider separating straight results and interpretation / discussion more clearly.
26. Please consider whether you can combine the key findings of figures 4-9 to achieve a synthesis that elevates these figures from data aggregation to interpretable results. Under the constraint of a very grainy image quality with generally illegible text inside the figures provided to this reviewer, figure 5 (day 1) seems to be by far the most informative time point, while differential expression seems to be almost absent by days 3 and 5 after first fever spike. It might be interesting to further probe this finding.
27. Lines 326-336 are rather basic / introductory and may be somewhat excessive for a reader with experience in the field.

Reviewer Confidential Comments to Editor:

As detailed in my summary and recommendation section in the author comments, this study is built on an interesting premise.
However, the execution is much to shallow to make for more than a mediocre concise report.

Based only on content and execution, I would recommend rejection with an invitation to resubmit after performing a significantly more thorough analysis of this dataset. However, I am convinced that all primary patient data should ultimately be published to inform meta-reviews; especially such hard to acquire data as early sepsis. I thus believe that this *can* be published as a data report if all review comments can be addressed.

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